

Philosophy behind Geometry

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Last updated: June 3, 2026

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✂ Introduction

This handout doesn't go really into formulas; rather, we delve into more of the general techniques you want to use when faced with very common problems. Hopefully with this in mind, this handout is more entry level for any beginners to learn from.

Unfortunately, this handout won't help you solve every geometry problem you face. You still, after all, need to know a ton of theory. But, it is just a good start especially for computational geometry contests. Lastly, this handout will be more problem-solving focused and we will attempt a lot of problems and see the philosophy at play later.

My ultimate goal with this handout is help guide you in your thought process as you proceed through many computational geometry problems. Although this obviously won't be a complete guide for every problem, I think it is a good start for your mentality at the very least.

※ Note on Diagrams

Two idea that I like to facilitate are

- “Make your diagram a work of art,” and
- “Label everything you know.”

In essence, your diagrams that you draw should be almost perfectly accurate and labeled. You want to be able to draw conclusions from your diagrams with ease. This doesn’t mean be a picasso, but make your diagrams sufficiently well drawn. Feel free to take multiple tries at making a diagram, but don’t let it take too long.

※ Note on Angle Chasing

In general, I like to say “When you don’t know what to do, just angle chase!” You will be awfully surprised as to how well this goes. Unfortunately, it isn’t so simple all the time. Here is a sample idea:

Identify any pairs of equal angles in the following setup.

Consider a triangle ABC and there exists a point P such that $AP = BP = CP$.

Observations. The above is a simple example. Note that equal sides means equal angles. For instance, $\triangle APB$ is isosceles with $AP = BP$. It therefore follows that $\angle PAB = \angle PBA$. Likewise, $\angle PBC = \angle PCB$ and $\angle PCA = \angle PAC$.

In this instance, it is important to notice equal sides implies equal angles. Now, not every problem is like this. You may have to notice some different ideas to obtain angle relationships. But, it all revolves around the key idea of using angles to establish relationships which will ultimately lead to fresh conclusions.

Example 1 (American Mathematics Competition). Let $ABCD$ be a square and E be a point outside square $ABCD$ such that ABE is an equilateral triangle. Compute the measure of $\angle AED$, in degrees.

Consider triangle AED : note that as $AD = AB$ due to square $ABCD$ and $AB = AE$ due to equilateral triangle ABE . Hence, it follows that $AE = AD$ and $\triangle ADE$ is isosceles. Angle chasing,

$$\angle DAE = \angle DAB + \angle BAE = 150^\circ.$$

It therefore follows that $\angle AED = \frac{180 - \angle DAE}{2} = 15^\circ$.

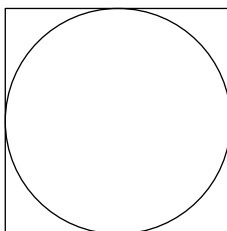
The key goal behind angle chasing is that it usually opens up many doors. By angle chasing, you will eventually see angle relationships, eventually possibly note similar triangles, note isosceles or equilateral triangles, or even cyclic quadrilaterals. This is a really good idea when you don’t know what to do and where to start.

※ Note on Circles

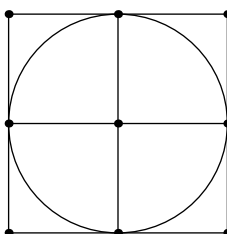
My greatest point on circles is “you know almost nothing about circles”. In particular, if you truly think about it, almost every formula you likely are aware of is related to lengths.

My strategy to offer you for circle problems is literally to just connect points together: centers of circles and tangencies, for instance. If you don't draw different segments then it is much harder to solve problems. Usually, you would be able to build connections by angle chasing or length chasing.

For instance, suppose I was given the following diagram:



Then, I would probably label all the tangencies and connect them with the center like so:



※ General Notes

When faced with geometry problems, usually you never want to sit there and stare at it. After all, it is really difficult to gain any insights. Instead, and this really goes for a bunch of topics in math: attack the problem. If you are faced with a problem, you want to do everything you can to solve it. And so, unless you've seen the same idea, you have to draw lengths or circles or possibly angle chase.

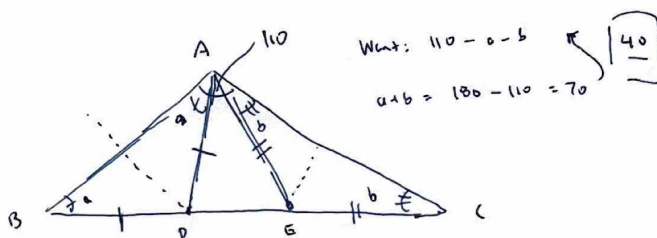
※ Some Examples

All of these problems which I will show you are from some high level math competitions. Despite that, they don't really require any advanced techniques. Just relying on the philosophy presented above and you should generally be fine. There may be a bit of synthetic observations you have to make here and there, though.

6.1 2025 EMCC

(2025 EMCC) Triangle ABC satisfies $\angle A = 110^\circ$. Points D and E lie on side $\overline{BC}$ such that $AB = BD$ and $AC = CE$. What is the measure of $\angle DAE$, in degrees?

Proof. Here is an example of some scratchwork I wrote:



From the condition,

$$\angle A = 110^\circ \implies \angle ABC + \angle ACB = 70^\circ.$$

As, $AD = DB$ and $AE = EC$ then triangle ADB and AEC are isosceles. Nicely enough the angle conditions $\angle ABC = \angle ABD = \angle BAD$ and $\angle ACB = \angle ACE = \angle CAE$ must hold. It follows that

$$\angle ABC + \angle ACB = 70^\circ \implies \angle BAD + \angle CAE = 70^\circ.$$

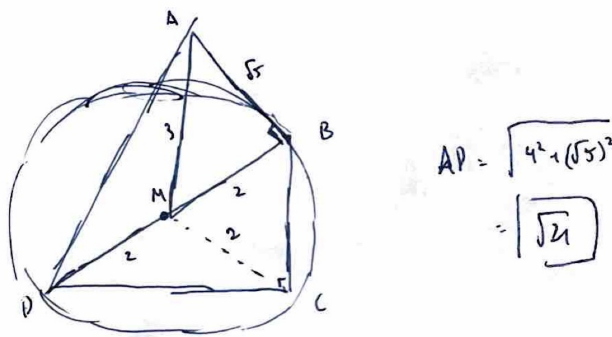
Therefore, $\angle DAE = \angle A - (\angle BAD + \angle CAE) = 110^\circ - 70^\circ = \boxed{40^\circ}$. Here are a few comments I'd like to point out from this:

- Notice, in the above diagram, it is clean and clear. You can see things marked clearly and it is always really helpful to have diagrams like these to make strong geometric observations and conclusions.
- For angle chasing, it may be helpful to set variables for unknown things. It is usually nice to be able to see some actual math going on like algebraic equations rather than a complex diagram.
- Always keep in the back of your mind what you are looking for. In this case, $\angle DAE = 110^\circ - a - b$. It is usually helpful to remind yourself of this to make sure you are always progressing to your final goal.
- Make sure to use all the conditions. In this case, we have tons of conditions like $\angle A = 110^\circ$ and also $AB = BD$ and $AC = CE$ which are key things to make sure to take advantage of. With this, it is helpful to make important geometric relationships like noticing $AD = DB$ implies an isosceles triangle and $\angle ABD = \angle DAB$.

6.2 2023 HMMT, Luke Robitaille and Milan Haiman

(2023 HMMT) Let $ABCD$ be a convex quadrilateral such that $\angle ABD = \angle BCD = 90^\circ$, and let M be the midpoint of segment BD . Suppose that $CM = 2$ and $AM = 3$. Compute AD .

Proof. Here is an example of some scratchwork I wrote:



Since M is the midpoint of BD and $\angle BCD$ is a right triangle, then BD is the diameter of the circumcircle of triangle BCD and so M is the circumcenter. It follows that $MB = MC = MD = 2$.

Due to Pythagoras, then $AB = \sqrt{AM^2 - MB^2} = \sqrt{5}$. So, again, due to Pythagoras:

$$AD = \sqrt{AB^2 + BD^2} = \sqrt{(\sqrt{5})^2 + 4^2} = \boxed{\sqrt{21}}.$$

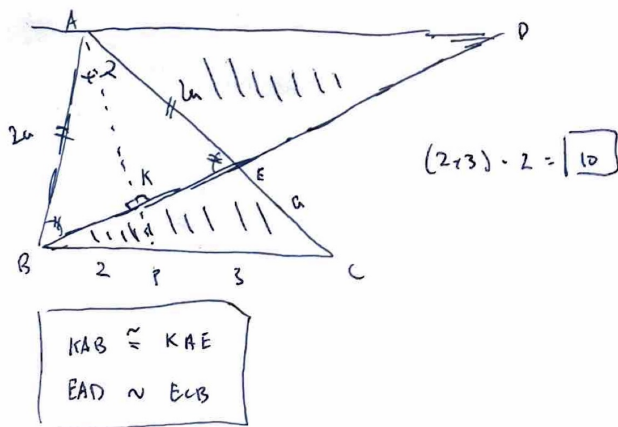
This problem isn't terribly difficult, but there are a few key things to take note of and we can learn from thanks to this problem:

- Firstly, a crystal clear diagram is always really helpful. Obviously, it isn't too perfect, but it is sufficient enough to make observations. Here, we were able to make relationships much easier.
- Try to make any synthetic observations. They may not work but sometimes they do and they can be helpful to get a better grasp anyways. In this case, making the connection that M is the circumcenter and so you can actually determine the length of MB and MD is a crucial first step. In addition, in the above instance, I draw the circumcircle of triangle BCD and noticed that the circumcircle is tangent to AB . Although this was never used, it is always helpful to make these connections just in case they may be used later on. Writing them down on the same could very well help.
- When you don't know what to do, it might be helpful in thinking of ways to eventually come up with the end result and tracking backwards. Taking advantage of the right angles and making the connection with Pythagoras is one good example for this problem.

6.3 2022 AMC, David Altizio

(2022 AMC, David Altizio) Let $\triangle ABC$ be a scalene triangle. Point P lies on $\overline{BC}$ so that $\overline{AP}$ bisects $\angle BAC$. The line through B perpendicular to $\overline{AP}$ intersects the line through A parallel to $\overline{BC}$ at point D . Suppose $BP = 2$ and $PC = 3$. What is AD ?

Proof. Here is an example of some scratchwork I wrote:



Let K be the intersection between AP and BD and E be the intersection between AC and BD . Due to the angle bisector theorem, denote $AB = 2a$ and $AC = 3a$ for some a .

As AK bisects $\angle BAE$ and $\angle BKA = \angle AKE = 90^\circ$, then $\triangle ABK \cong \triangle AEK$. Therefore, we may conclude that $AE = AB = 2a$ and so we obtain that $CE = a$.

Notice, $\angle EAD = \angle ECB$ and $\angle EBC = \angle EDA$ so $\triangle ECB \sim \triangle EAD$. From ratios between the two triangles:

$$\frac{EC}{CB} = \frac{EA}{AD} \implies \frac{a}{5} = \frac{2a}{AD} \implies AD = \boxed{10}.$$

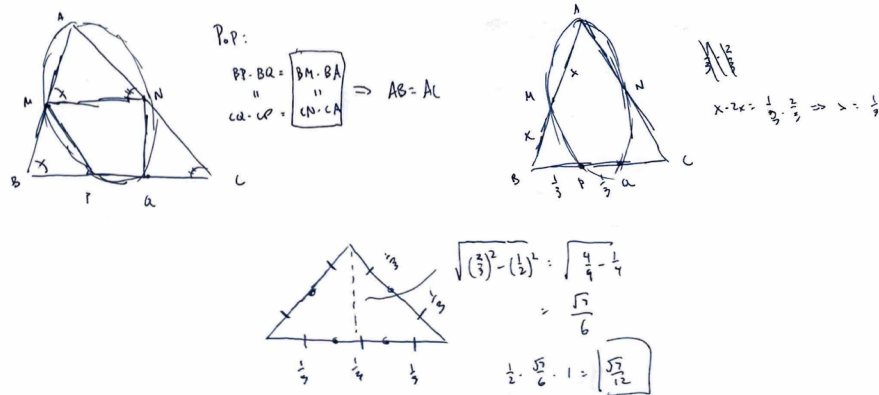
Fun fact, I recall coming up with a lot of the ideas in the “angle chasing” section prior to the this exam. I think this was a pretty memorable moment for me because it is the first time I felt I kinda applied this idea, and it worked! Irrelevantly, the first two problems stress a lot of points already, so the next summary ideas will be shorter:

- Angle chase when you don’t know what to do! As I said above, this is a big problem where noticing a lot of relationships will lead you to the pair of congruent triangles and the pair of similar triangles.
- Consider keeping a list of observations or conclusions you came up with. As you can see above, I ended up boxing the observations I made which help me remind myself of what I had come up with already.
- Synthetic observations were also key here in recalling angle bisector theorem and using equal lengths and such. It is important to always somehow take advantage of each of these when you see them.

6.4 2024 HMMT, Rishabh Das

(2024 HMMT) In triangle ABC , points M and N are the midpoints of AB and AC , respectively, and points P and Q trisect BC . Given that A, M, P, N , and Q lie on a circle and $BC = 1$, compute the area of triangle ABC .

Proof. Here is an example of some scratchwork I wrote:



From power of the point at B and C , then

$$\begin{aligned} BM \cdot BA &= BP \cdot BQ \implies 2 \cdot BM^2 = 2 \cdot BP^2 \implies BM = BP, \\ CN \cdot CA &= CQ \cdot CP \implies 2 \cdot CN^2 = 2 \cdot CQ^2 \implies CN = CQ. \end{aligned}$$

Recall, $BP = CQ$ and so actually $BM = CN$. Hence,

$$AM = MB = BP = PQ = QC = CN = CA = \frac{1}{3}.$$

The height from A to BC can be found as

$$d(A, BC) = \sqrt{AC^2 - \left(\frac{BC}{2}\right)^2} = \sqrt{\left(\frac{2}{3}\right)^2 - \left(\frac{1}{2}\right)^2} = \frac{\sqrt{7}}{6}.$$

$$\text{Hence, } [ABC] = \frac{d(A, BC) \cdot BC}{2} = \boxed{\frac{\sqrt{7}}{12}}.$$

This problem has a lot of steps, but a lot of them are generally repeats of what we discussed before. I just want to emphasize a few unique items for this problem:

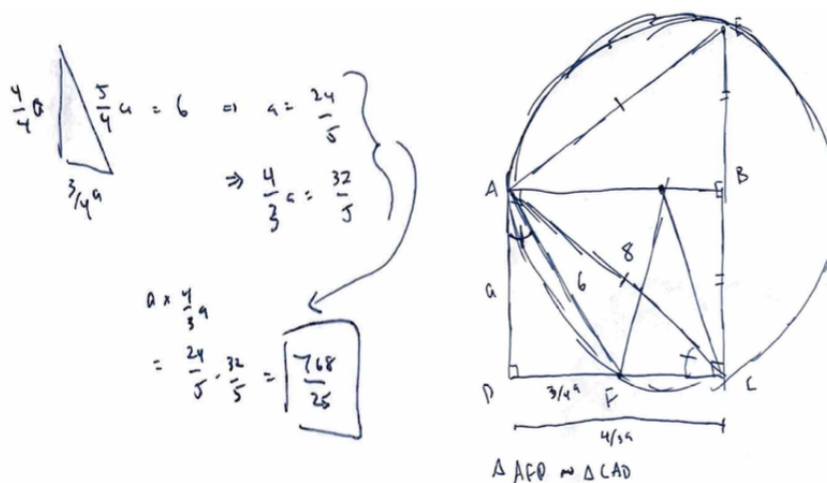
- Feel free to take multiple attempts at drawing the diagram as you gather more information. As you can see, it took three tries for me to get a diagram and enough information.
- Some things won't work all the time. As you can see in the first diagram, I tried to start an angle chase, but it didn't work too well and there didn't seem to be enough to actually deal with. Despite that, it is good to take a look back and see if you can do anything with the lengths and it turns out the conditions given usually deal with the lengths of the figure.

6.5 2026 HMMT, Rishabh Das

Here is a really good example. I solved this back during the Guts Round when I was part of the Florida Math Team, so let's give it a try together.

(2026 HMMT) Let $ABCD$ be a rectangle. Let E be the reflection of C over B . The circumcircle of triangle ACE intersects line CD at a point $F \neq C$. Given that $AC = 8$ and $AF = 6$, compute the area of rectangle $ABCD$.

Proof. Here is an example of some scratchwork I wrote:



Note that $BE = BC$ and $EA = AC$. Hence, it follows the triangle is an isosceles triangle. So, the circumcenter of AEC lies on AB , the perpendicular bisector of CE . Therefore,

$$\angle ACD = \angle FAD \implies \triangle ACD \sim \triangle FAD.$$

Hence, $AD : DF = 4 : 3$, and so $AD = \frac{4}{5} \cdot AF = \frac{24}{5}$. Then, $DC = \frac{32}{5}$. The area, therefore, is

$$AD \cdot DC = \frac{24}{5} \cdot \frac{32}{5} = \boxed{\frac{768}{25}}.$$

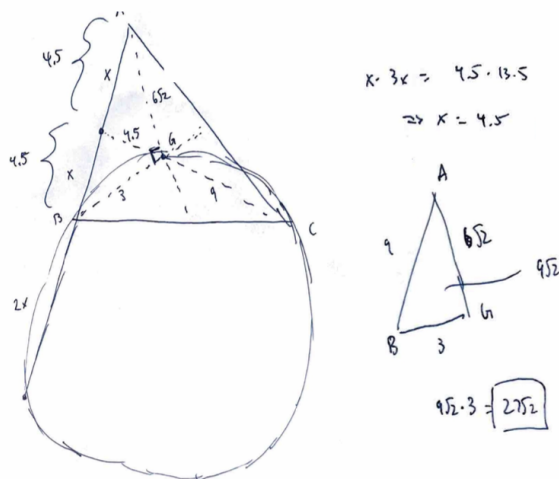
Once again, a lot of points are repeats. However, there are a few key points to shout out here and so here they are:

- I strongly recall just angle chasing and the pieces magically falling together when I did this in-test. If I wasn't aware of the idea to look at angles, I am definitely sure this would've taken a ton more time or maybe I wouldn't have solved it at all. You can also see how we used the angle chasing to lead us to length chasing here, as well.
- There is a ton of synthetic ideas that are super subtle but super important. I recommend going through them and trying to motivate each one.

6.6 2025 FLSAM, Jacob Khohayting

Let ABC be a triangle with centroid G . Suppose the reflection of A across B lies on the circumcircle of BGC . Given that $GB = 3$ and $GC = 9$, find the area of ABC .

Proof. Here is an example of some scratchwork I wrote:



Let M be the midpoint of AB and X be the reflection of A over B . From power of a point, it therefore follows that:

$$MB \cdot MX = MG \cdot MC \implies MB \cdot (3 \cdot MB) = MG \cdot (3 \cdot MG) \implies MB = MG = 4.5.$$

Hence, $MA = MB = MG$ and so M is the circumcenter of ABG and $\angle AGB = 90^\circ$. Now, $AB = 9$ and $BG = 3$ and so from Pythagoras, $AG = 6\sqrt{2}$. Computing the area, then $[ABG] = 9\sqrt{2}$. From centroid properties,

$$[ABC] = 3 \cdot [ABG] = 3 \cdot 9\sqrt{2} = \boxed{27\sqrt{2}}.$$

I don't actually have to many points for these problems. In particular, a lot of it is using synthetic observations and making conclusions from those. However, I do think this actually has a sole point I want to show you which is the following:

- There are very few things you know about circles, as I stated in my previous section on "Notes on Circles". Power of a point, at the end of the day, is generally a formula involving lengths. And so, it is extremely crucial to pay attention to the lengths involved in this problem rather than the circle itself.